EET202

HINES & TRANSFORMERS

MODULE: 4

– Equivalent Circuit, Losses and Efficiency

Prepared By,

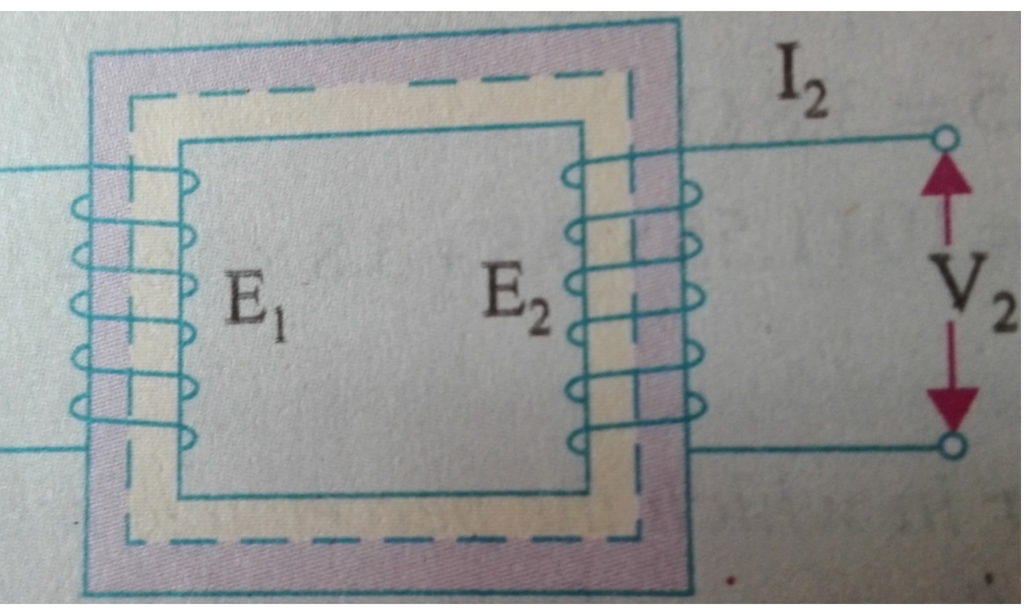
ADARSH SR
Asst. Professor
EEE Department
SCET Kodakara

Syllabus

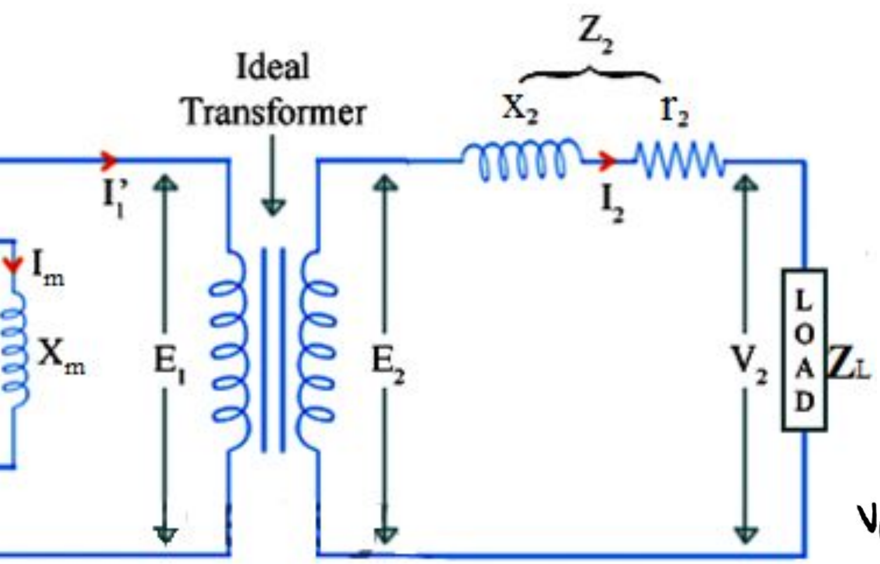
ers –constructional details, principle of operation, EMF equation,
convention, magnetising current, transformation ratio, phasor
no load and on load, equivalent circuit, percentage and per unit
egulation. Transformer losses and efficiency, condition for
VA rating. Testing of transformers– polarity test, open circuit test,
npner's test – separation of losses, all day efficiency.Parallel
se transformers– numerical problems

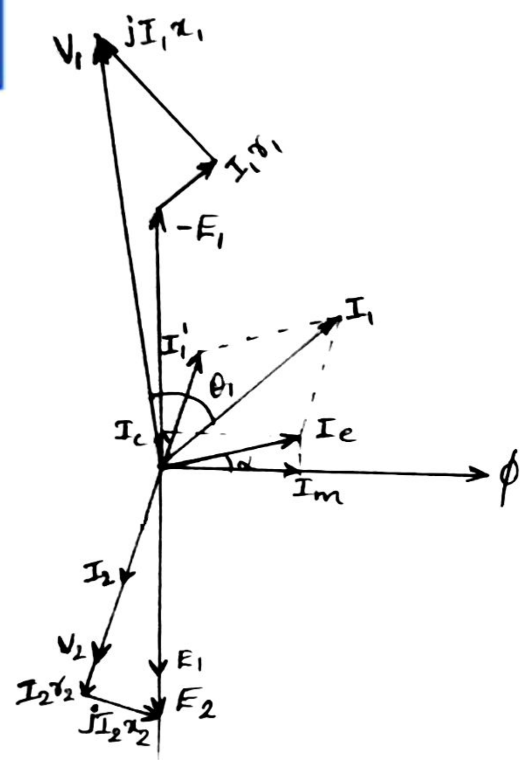
Equivalent Circuit

t of a machine means, the circuit which exactly
rformance and working of the machine.

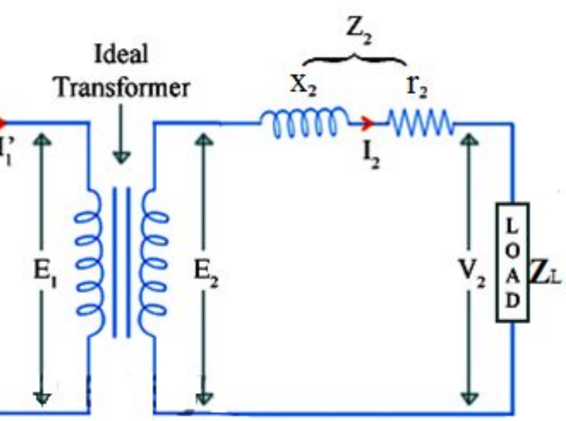


Equivalent Circuit





alent Circuit referred to Primary

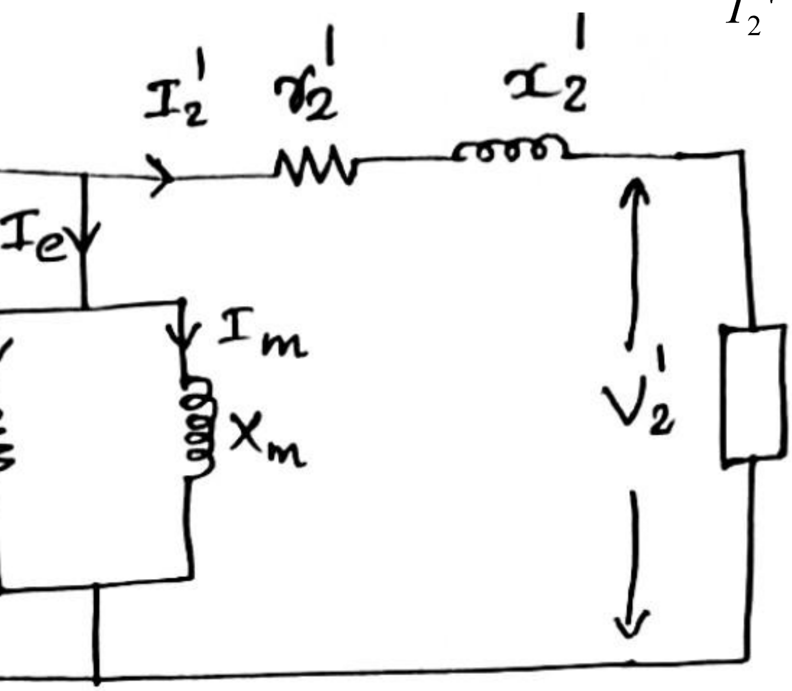


$$\frac{V_2}{V_1} = K = \frac{I_1}{I_2}$$

$$V_2' = \frac{V_2}{K}$$

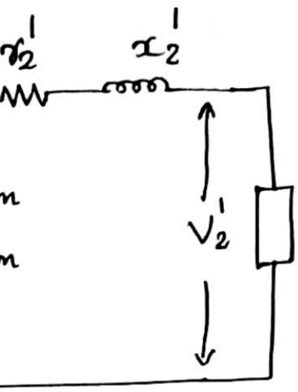
$$I_2' = KI_2$$

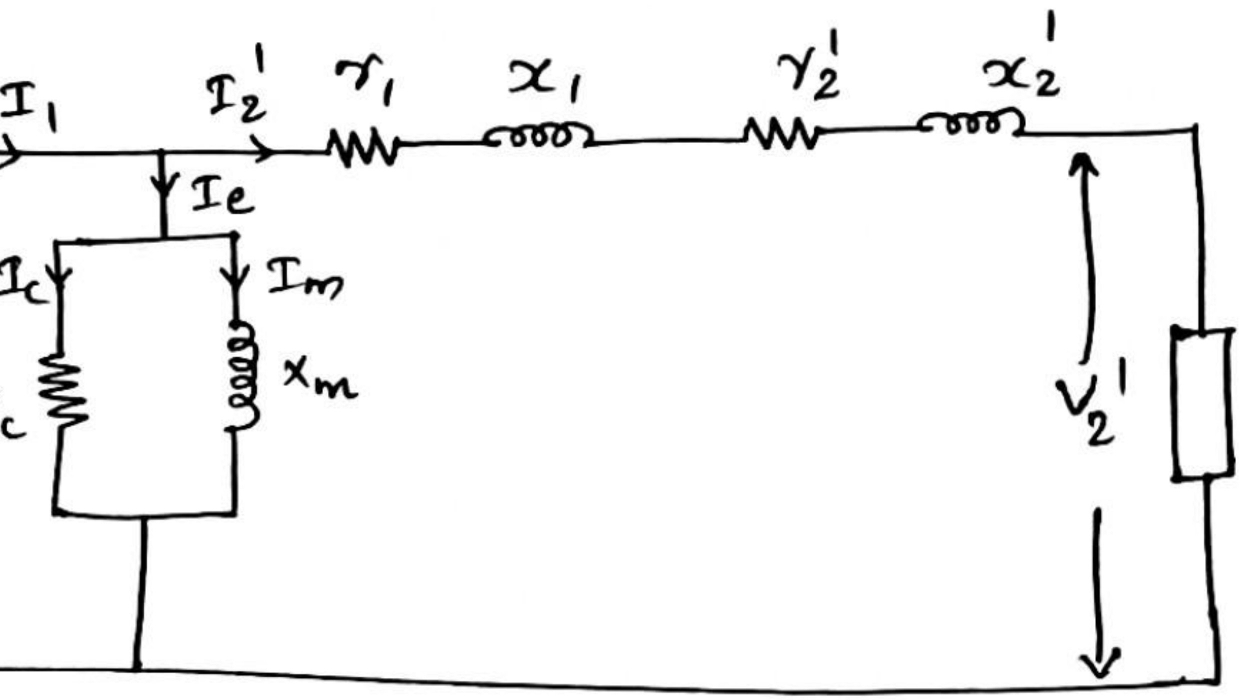
$$Z_2' = \frac{V_2'}{I_2'} = \frac{(V_2 / K)}{KI_2} = \frac{1}{K^2}\frac{V_2}{I_2} = \frac{Z_2}{K^2}$$



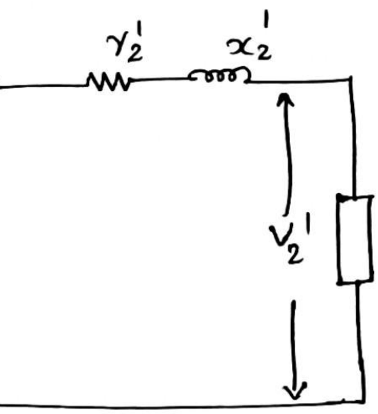
$$\boxed{r_2' = \frac{r_2}{K^2} \qquad x_2' = \frac{x_2}{K^2}}$$

ximate Equivalent Circuit





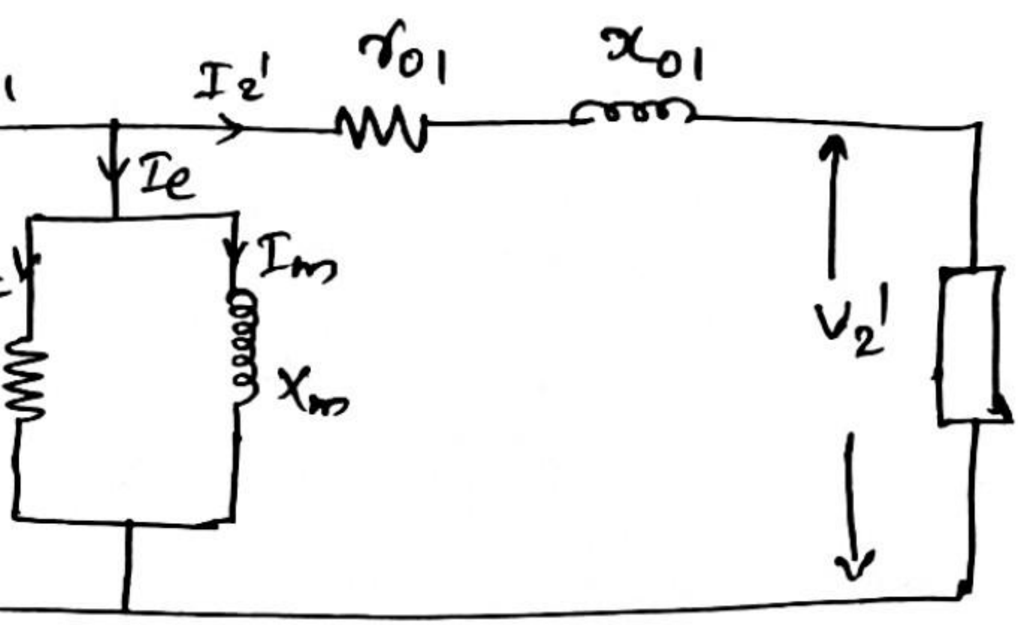
...te Equivalent Circuit referred to Primary



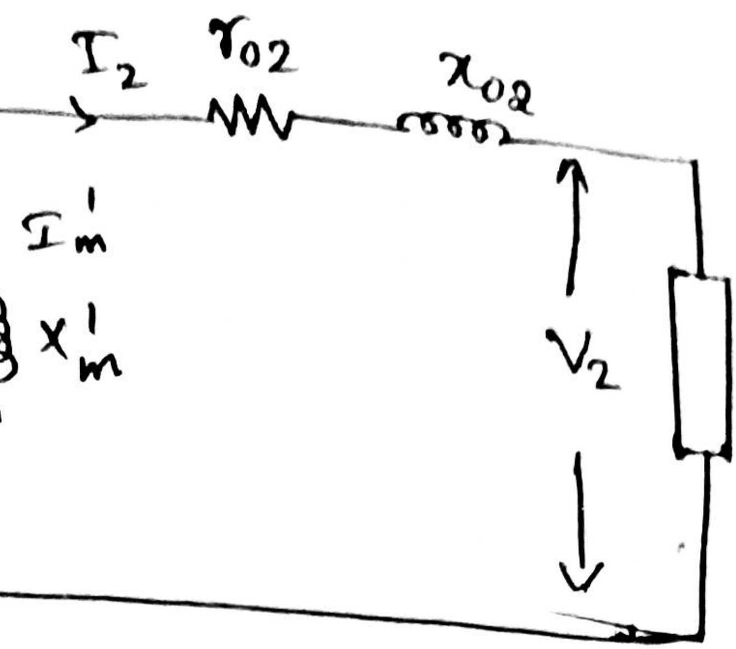
$$r_{01} = r_1 + r_2'$$

$$x_{01} = x_1 + x_2'$$

$$z_{01} = \sqrt{r_{01}^{2} + x_{01}^{2}}$$



te Equivalent Circuit referred to Secondary



$r_1' = K^2 r_1$

$x_1' = K^2 x_2$

$X_m' = K^2 X_m$

$R_c' = K^2 R_c$

$' + r_2$

$_1' + x_2$

$\sqrt{r_{02}^2 + x_{02}^2}$

osses and Efficiency

d with a transformer are:

ron Losses)

s

ut / Input = (Input – Losses) / Input

on for Maximum Efficiency

$$= I_2^{\,2} r_{02}$$

$$\frac{utput}{nput} = \frac{Input - Losses}{Input}$$

$$\frac{I_1 \cos\phi_1 - I_1^{\,2} r_{01} - W_i}{V_1 I_1 \cos\phi_1}$$

$$\frac{I_1 \cos\phi_1}{I_1 \cos\phi_1} - \frac{I_1^{\,2} r_{01}}{V_1 I_1 \cos\phi_1} - \frac{W_i}{V_1 I_1 \cos\phi_1}$$

$$- \frac{I_1 r_{01}}{V_1 \cos\phi_1} - \frac{W_i}{V_1 I_1 \cos\phi_1}$$

n for Maximum Efficiency

n respect to I_1

$$\frac{d\eta}{dI_1} = 0 - \frac{r_{01}}{V_1 \cos\phi_1} + \frac{W_i}{V_1 I_1^2 \cos\phi_1}$$

iency, $\frac{d\eta}{dI_1} = 0$

$$\frac{r_{01}}{V_1 \cos\phi_1} = \frac{W_i}{V_1 I_1^2 \cos\phi_1}$$

$$\boxed{I_1^2 r_{01} = W_i}$$

$$\boxed{Cu\ loss = Core\ loss}$$